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3.1 | Deck Design Properties

Import deck loads and functions.

Table 1: Load values from rv102-loads.py (rv_stor/v102-2.csv)

variable	value	[value]	description
D_1	2.00 plf	0.00 klf	2x6 planks DL
D_2	2.60 plf	0.00 klf	2x8 joists DL
D_3	2.90 plf	0.00 klf	4x4 posts and struts
E_1	29000.00 ksi	199947.96 MPA	modulus of elasticity
LL_1	40.00 psf	1.92 kPA	ASCE7-05 floor LL
HL_1	20.00 psf	0.96 kPA	ASCE7-05 HL

Table 2: Import Functions (rvsrc/scripts/checks.py)

Function	Docstring
nds_beam_check(** kwargs)	Check stress and deflection for a simply supported wood beam using NDS.
nds_post_check(** kwargs)	Check stress at cantilever post

3.1 - 2 | Deck Design Summary

Design properties as dictionary for checking function nds_beam_chk

```
Function Arguments Dictionary : beam1 (units: inch, pounds)
=====
ln_1 = 4*12. # beam span
w_1 = 45*.5/12 # uniform linear load
b_1 = 5.5 # beam width
d_1 = 1.5 # beam depth
E_1 = 1.5*(10**6) # modulus of elasticity
F_b = 1000 # allowable bending stress
C_D = 1.0 # load duration factor
C_M = 0.85 # wet service factor
C_F = 1.0 # size factor
C_t = 1.0 # temperature factor
C_i = 0.8 # incising factor
C_r = 1.0 # repetitive member factor
C_c = 1.0 # curvature factor
C_L = 1.0 # beam stability factor
C_b = 1.0 # bearing area factor
deflect_limit = 240.0 # max allowable deflection ln_1/deflect_limit
=====
```



Design Results

Eq. 1: Check Deck Beam

```
function: nds_beam_check | args: beam1 | units: inch, pounds

Beam Check Results:
=====
total UDL: 1.88
fb: 261.82
Fb_prime: 680.00
E_prime: 1275000.00
stress_ratio: 0.39
deflection: 0.07
deflection_ratio: 0.20
```

3.1 - 3 | Strut

Check strut D/C ratio with BeamChek 2023

BeamChek v2023 licensed to: StL Reg # 2999-68413						
Tree Fort		Struts				
		Prepared by:			Date: 6/09/26	
<u>Selection</u>	4x 4 DF-L Select Solid Wood Column					
<u>Conditions</u>	NDS 2018, Using values for 2x and 4x solid sawn, Dimension Lumber.					
	Wet Use					
<u>Data</u>	Load	1100 #	Column Area	12.25 in²	Kf	1.00
	Actual Height	8.0 ft	le d1 Effective Ht	96 in	c	0.80
	Unbraced L1	8.0 ft	le d2 Effective Ht	96 in	KcE	0.30
	Unbraced L2	8.0 ft	Ke Buckling Mode	1.0	FcE	683
<u>Attributes and Values</u>	Controlling d is 3.5 inches			Fc (psi) E (psi x mil)		
	le/d	psi	Area (in²)	Reference Values	1700	1.9
				Adjusted Values	606	1.7
	Actual	27	90	CF Size Factor	1.15	
	Critical	50	606	Cd Duration	1.00	
	Status	OK	OK	Cm Wet Use	0.80	0.90
	Ratio	54%	15%	Cp Stability	0.39	
Note: A wood plate under this column must have an Fc value, perpendicular to the grain, greater than 90 psi.						

Fig. 1 - Strut Check